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SumoRobot

by

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Mechatronic Project 478

Final Report

Study leader: Mnr W. Swart

November 2026

Verklaring

Ek, die ondergetekende, verklaar hiermee dat die werk in hierdie verslag vervat, my eie oorspronklike werk is.

Handtekening: .....
S. Bekker

Datum: .....

EXECUTIVE SUMMARY

Title of Project
Design and Development of an Autonomous Mini Sumo Robot.
Objectives
Design and construct an autonomous Mini Sumo robot capable of detecting, pushing and removing an opponent from the sumo arena while remaining within the competition boundaries and complying with all competition constraints.
What is current practice and what are its limitations?
Current Mini Sumo robots primarily rely on simple sensor-based control systems for line detection and opponent detection. Their performance is often limited by restricted adaptability to different opponents, inefficient pushing strategies and the strict size, weight and power limitations imposed by competition rules.
What is new in this project?
This project investigates improved control strategies, sensor integration and mechanical design to enhance the robot's ability to detect opponents, remain within the arena and generate sufficient traction and pushing force during competition.
If the project is successful, how will it make a difference?
The completed robot will demonstrate a balanced integration of sensing, control and mechanical design for competitive Mini Sumo robotics. The resulting platform, control strategies and design methodology can serve as a reference for future robotics competitions and educational robotics projects. Several subsystems, including the control software and sensor integration techniques, can also be adapted for use in other mobile robotic applications.
What are the risks to the project being a success? Why is it expected to be successful?
Potential risks include unreliable sensor measurements, insufficient wheel traction or pushing force, and software or control issues affecting the robot's behaviour. These risks are mitigated through iterative testing, careful component selection, simulation, prototype evaluation and systematic software development throughout the project.
What contributions have/will other students make?
None. The project is conducted independently and does not directly build upon previous university robotics projects.
Which aspects of the project will carry on after completion and why?
The robot platform and control software can be further developed to improve autonomous strategies, sensing capabilities and mechanical performance for future robotics competitions or research projects.
What arrangements have been/will be made to expedite continuation?
Comprehensive documentation of the design process, hardware selection, software implementation, testing procedures and experimental results will be produced to assist future students in extending or improving the project.

Abstract

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Nomenklatuur

Konstantes

g =	$9.81\,\mathrm{m/s^2}$	

Veranderlikes

Re_{D}	Reynolds getal t.o.v. deursnit	[ ]
x	Koordinaat	[m]
$\ddot{x}$	Versnelling	[$\mathrm{m/s^2}$]
θ	Rotasiehoek	[rad]
τ	Moment	[N·m]

Vektore en tensore

$\overrightarrow{\boldsymbol{v}}$	Fisiese vektor, sien vergelyking ...

Subscripts

a	Adiabaties
a	Koordinaat

Chapter 1

Introduction

1.1 Background

Robotics competitions form an environment where participants can compete against each other - showcasing their designs and competitive strategies. Sumorobot competitions are a well-known example. Robots are required to navigate the arena and fight against the opposing robot by pushing them out of the boundaries of the ring.

This project focuses specifically on the Autonomous Mini Sumobot class. A robot must be designed and developed to comply with the Robogames competition rules. The robot is expected to use sensors and control/strategy algorithms to detect said opponent, navigate the arena and then apply enough force to counter the opposing robot and push it out of the arena bounds. The most important components of the robot design being speed, momentum, pushing force, traction and precise autonomous strategies.

The project combines knowledge from multiple different engineering tools. The main module knowledge used for this project stems from electronics, computer systems and machine design as well as key skills developed during the E-design module. By combining mechanical design, electronic systems and control strategies, the project will aim to create a functional and competitive robot. The completed robot will then be used in a knock-out style match between students to evaluate its performance.

1.2 Objectives

The objective of this project is to design and develop a compact autonomous Mini Sumo robot that complies with the Robogames Mini Sumo competition regula-

tions. The robot must be capable of detecting opponents while remaining within the arena boundaries through the implementation of a sensing and feedback system. A suitable control strategy will be developed to enable the robot to autonomously locate, engage and push its opponent out of the arena while avoiding the boundary line. Finally, the completed system will be validated through experimental testing and participation in the Mini Sumo knock-out competition to demonstrate its functionality and overall performance.

1.3 Motivation

Autonomous robotics combines theoretical knowledge, design and testing in a loop of trial, error and improvement. Such a seemingly simple project requires knowledge of a variety of engineering principles. It integrates mechanical design, the design of electronic systems and the implementation of control and strategy algorithms.

All the above makes the Sumobot project particularly valuable for developing practical engineering and problem-solving skills. The design of a Mini Sumo robot requires careful consideration of traction, stability, sensing capabilities and control strategies to detect an opponent and successfully shove it out of the arena while remaining within the ring boundaries.

This project makes use of knowledge gained in the Computer systems, Electronics, Control systems, E-design and Machine Design modules. By integrating these areas, the project provides an opportunity to demonstrate the practical application of engineering principles in the design and development of a autonomous robotic system.

Chapter 2

Literature Review

2.1 Competition Requirements and Design Constraints

The design of a mini sumobot is mainly limited by the competition guidelines and rules. This includes the physical dimensions of the robot, the characteristics and layout of the competition arena and the operational requirements and limits that must be satisfied during a match. The following section provides an overview of the regulations and rules according to Robogames. The Mini sumobot class is restricted to a maximum footprint of 100 mm x 100 mm with no height limit (RoboGames, 2026). The weight is limited to 500 g as well and the robot must be fully autonomous, with no remote control or human intervention allowed during a match.

The Robogames mini sumo competition arena, also known as the dohyo, is a circular platform with a diameter of 77 cm and a height of 2.5 cm. The surface is made of a smooth, black melamine coated wood and the boundary is marked by a white border with a width of 2.5 cm as seen in Figure 2.1.1. Inside this arena, two centerlines (also called shikiri-lines) are drawn to indicate the starting positions of the robots. The robots can be placed in any position behind the shikri-line (RoboGames, 2026).

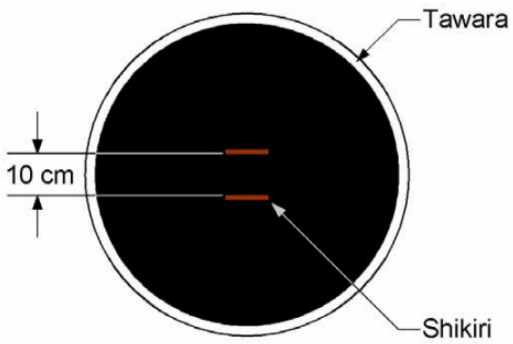


Figure 2.1.1: Standard mini sumo competition arena (Parallax Inc., 2005).

The main objective of these robots is to push the opponent out of the dohyo while avoiding being pushed out themselves. A 'Yuko' is scored when a robot is pushed out of the dohyo and the first robot to score two 'Yuko' wins the match. If both robots are pushed out of the dohyo simultaneously, the match is declared a draw and must be restarted (RoboGames, 2026). It should be noted that the use of any jammers or devices that could cause harm to the opponent is strictly prohibited. Using such devices could result in disqualification from the competition.

2.2 Control Strategy

The control strategy of sumorobots can ultimately be reduced to a simple state machine with four primary operating states:

- Start: waiting for the competition start signal and allowing the required delay period of 5 seconds (RoboGames, 2026).
- Search: rotating or moving to locate the opponent.
- Attack: driving towards the opponent while maintaining forward force.
- Back-off: reversing or turning when the dohyo boundary is detected.

Most sumobot hobbyists do not make use of closed-loop control, but rather use a simple open-loop control strategy. However, it can be useful to consider some form of PID control given the impact the opponent's behaviour might have on the robot's performance.

2.3 Common Sumobot Sensors

The first step after starting the robot is to locate the opponent. Various sensor types can be used to detect the opponent, including infrared, ultrasonic, time-of-flight and vision-based sensors. The choice of sensor depends on factors such as detection range, response time, size, weight and power consumption.

Ultrasonic sensors measure distance by emitting a pulse of sound and measuring the time it takes for the echo to return. They are effective for detecting objects at medium range but may struggle with small or angled surfaces (Santos, 2013).

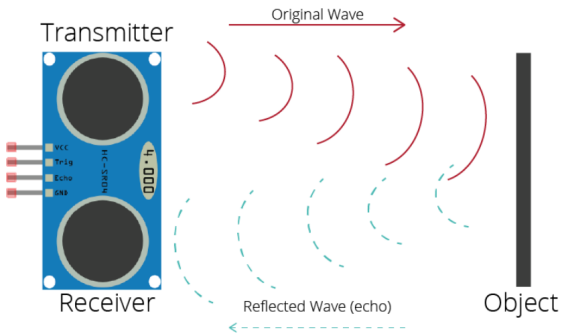


Figure 2.3.1: HC-SR04 ultrasonic sensor (Santos, 2013).

With the uncertainty of the opponent's design and possible need for accurately detecting angled surfaces, time-of-flight sensors may be a better option and are frequently used by competitive sumo robot teams. Time of flight sensors operate by emitting a light pulse and measuring the time it takes for the reflection to return. They are known for their fast response time and accuracy under different light conditions (ToF Sensors, 2026).

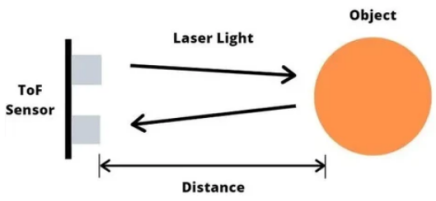


Figure 2.3.2: Time-of-Flight sensor (ToF Sensors, 2026).

IR sensors are used for opponent detection, but they can't measure distance as accurately as the other options. They work on the basis of detecting that there is something within its sensing range, but it cannot determine how far away the object is (Sisko, 2017).

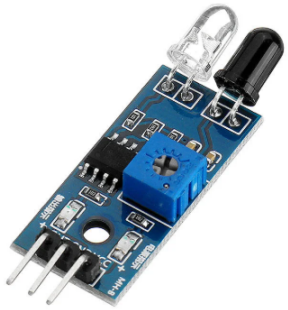


Figure 2.3.3: IR sensor (Electrobes, 2026).

2.4 Communication and Sensor Interfaces

I2C, register allocation and SPI communication protocols will be discussed here.

2.5 Differential Drive Kinematics

Differential drive robots have two actuated wheels that operate on the same axis. These robots can be summarized by a state description consisting of both orientation and position (California Institute of Technology, 2018). This state can be modeled within the x-y plane as shown in Figure 2.5.1.

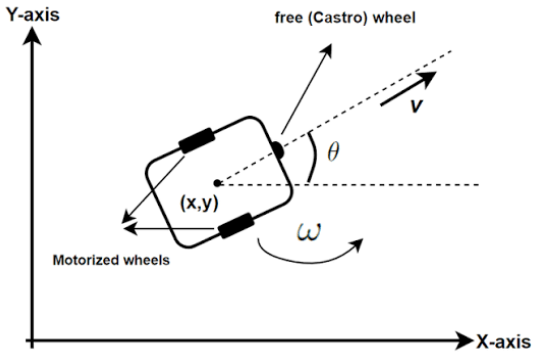


Figure 2.5.1: Differential drive robot kinematics (California Institute of Technology, 2018).

This figure represents a differential drive robot commonly used in 'Duckie Town' which includes a free castor wheel. However, these equations still apply to a differential drive robot without a castor wheel.

Assume the action vector of the robot is represented by $u = [u_r, u_l]$, where u_l and u_r represent the left and right wheel velocities respectively. When both wheels rotate at the same angular velocity ($u_l = u_r > 0$), the differential-drive robot moves in a straight line along its current heading. The resulting linear velocity is directly proportional to the wheel radius, r. When the wheels rotate at equal speeds in opposite directions ($u_l = -u_r \neq 0$), the robot undergoes rotational motion about the midpoint of the axis of the wheels, producing no translational movement. From there on, the following equations can be used to determine the robot's position and orientation over time:

$$\dot{x} = \frac{r}{2}(u_r + u_l)cos(\theta) \tag{2.5.1}$$

$$\dot{y} = \frac{r}{2}(u_r + u_l)sin(\theta) \tag{2.5.2}$$

$$\dot{\theta} = \frac{r}{L}(u_r - u_l) \tag{2.5.3}$$

Where (x, y) is the position of the robot in the x-y plane, θ is the orientation of the robot, and L is the distance between the two wheels (Ras, 2025).

2.6 Wheel Material

2.6.1 Polyurethane

Polyurethane and its density as well as friction factor and shore rating will be discussed here (what shore ratings are)

2.6.2 Silicone Rubber

2.7 Power Supply

2.7.1 Lithium Polymer (LiPo) Batteries

2.7.2 Lithium Ion (Li-ion) Batteries

2.7.3 Alkaline Batteries

Chapter 3

Specifications and Requirements

3.1 Stakeholder Requirements

The stakeholder requirements for the autonomous sumorobot are listed in Table 3.1.1. The requirements are ranked in order of importance, with the most important requirement being ranked first. The primary stakeholders for this project include that staff and students who will be directly involved in the manufacturing, testing and the competition of the robot. The requirements are therefore based on the needs of these stakeholders, as well as the competition rules and regulations.

Table 3.1.1: Stakeholder requirements

SR ID	Description	Rank
SR1	The robot must comply with all Mini Sumo competition regulations (size, mass and safety regulations).	i
SR2	The robot must operate autonomously during the match.	ii
SR3	The robot must remain within the dohyo boundary.	iii
SR4	The robot must have enough traction and pushing force to compete against an opponent.	iv
SR5	The robot must withstand impacts or collisions and pushing forces without mechanical or electrical failure.	v
SR6	The robot must operate reliably throughout testing and competition without unexpected resets or system-related failures.	vi
SR7	The robot must complete a full match using its contained power supply.	vii

Continued on next page

Table 3.1.1 continued from previous page

SR ID	Description	Rank
SR8	The robot must be manufacturable using the available facilities, components and project budget.	viii

3.2 Functional Requirements

The functional requirements are derived from the stakeholder requirements and are listed in Table 3.2.1 below.

Table 3.2.1: Functional requirements for the autonomous sumorobot

FR ID	Description	Derived from
FR_01	The robot shall detect the opposing robot.	SR2
FR_02	The robot shall detect the edge of the dohyo.	SR3
FR_03	The robot shall move within the dohyo.	SR2
FR_04	The robot shall apply sufficient force to push the opposing robot out of the dohyo.	SR4
FR_05	The robot shall respond to the opposing robot in real time.	SR2, SR6
FR_06	The robot shall operate autonomously without human intervention after the start command.	SR2, SR3
FR_07	The robot shall remain within the size and mass limits specified for the competition class.	SR1
FR_08	The robot shall be self-contained and shall not rely on external power, communication or physical support during a match.	SR1, SR7
FR_09	The robot shall avoid the dohyo edge and shall prioritise boundary avoidance over attacking the opposing robot.	SR3, SR6
FR_10	The robot shall begin operation only after the required competition start delay.	SR1
FR_11	The robot shall not damage the dohyo, the opposing robot or itself during normal operation.	SR1, SR5

Continued on next page

Table 3.2.1 continued from previous page

FR ID	Description	Derived from
FR_12	The robot shall incorporate components that can be manufactured using the available manufacturing and workshop facilities.	SR8
FR_13	The robot shall be manufactured from materials and designed in such a way that it can withstand the forces and impacts experienced during a match.	SR5

3.3 Technical Performance Measures

The technical performance measures (TPMs) are derived from the functional requirements and are listed in Table 3.3.1. These performance measures will be used to evaluate and test the system.

Table 3.3.1: Technical performance measures for the autonomous sumorobot

TPM ID	Related FR	Description	Target	Range	Unit
TPM_01	FR_06	Start delay after power-on[1].	5	5–10	s
TPM_02	FR_01	Opponent detection time[8].	–	0–100	ms
TPM_03	FR_07	Robot footprint[2].	100×100	$\leq 100 \times 100$	mm
TPM_04	FR_07	Robot mass[2].	500	≤ 500	g
TPM_05	FR_07	Robot height[2].	40	< 100	mm
TPM_06	FR_02	Dohyo edge detection time[8].	50	0–100	ms
TPM_07	FR_03	Maximum forward speed[9].	0.30	0.10–0.77	m/s
TPM_08	FR_04	Available wheel torque[3].	1.5	0.5–1.5	kgf·cm
TPM_09	FR_08	Nominal battery voltage[4].	11.1	7.4–12.6	V
TPM_10	FR_05	Overall sensing latency.	100	0–100	ms
TPM_11	FR_05	Robot response time after detection.	150	0–200	ms

Continued on next page

Table 3.3.1 continued from previous page

TPM ID	Related FR	Description	Target	Range	Unit
TPM_12	FR_03	Turning radius.	50	0–50	mm
TPM_13	FR_01	Opponent detection range[5].	70	0–77	mm
TPM_14	FR_01	Opponent detection field of view[5].	60	40–180	°
TPM_15	FR_02	Edge detection distance from the boundary.	3	0–5	mm
TPM_16	FR_02	Edge avoidance reaction time.	100	0–200	ms
TPM_17	FR_03	Time to reach operating speed.	1	0.5–2	s
TPM_18	FR_04	Continuous pushing duration[7].	5	2–10	s
TPM_19	FR_05	Control-loop update rate.	20	10–100	Hz
TPM_20	FR_06	Successful autonomous runs.	8/10	0–10	runs
TPM_21	FR_08	Battery voltage under load[4].	10.5	9–12.6	V
TPM_22	FR_08	Continuous operating time[6].	12	3–24	min

[1] According to the Robogames Mini Sumo competition rules.
[2] Maximum robot dimensions and mass are specified by the Robogames Mini Sumo competition regulations.
[3] Wheel torque is specified per drive motor at the gearbox output shaft (see Appendix B).
[4] Values are based on the voltage requirements of most Sumobot peripherals.
[5] Values are based on ideal sensor performance and ranges.
[6] Matches last for 3 minutes maximum according to the Robogames Mini Sumo competition rules. Ideally, the robot should be able to run for 3 consecutive matches without recharging or replacing the power source.
[7] Continuous pushing duration is based on the average time it takes to push an opponent out of the dohyo or to overcome wheel slip. The motor should not stall within this time.
[8] Estimate sensor processing time based on typical operation ratings for various sensors.
[9] Reaching this speed would allow the robot to cross roughly half of the dohyo in a second.

Chapter 4

Concept Design

PLEASE NOTE: This chapter is far longer than my goal at the moment. It will be shortened and some parts will either be moved to or merged with the final design. Currently, these sections are simply here to document work done and to keep track of the desicion-making process.

Based on pre-existing designs, the design of the autonomous sumorobot can be broken down into mechanical and electrical components. The mechanical components include the chassis, drive train, wheels, wedge and supporting frame which allows it to interface with the electrical components. The electrical components include the microcontroller, sensors, motor drivers and power supply which allow the robot to detect its environment and control its movement. This chapter will discuss the concept design of these various sub-components.

4.1 Mechanical Concept Designs

The diagram below outlines the mechanical sub-components of interest that will be discussed in this section. The supporting frame will not be discussed in detail as it can only be designed in the refinement stage once electrical components and their positioning have been selected. The drive-train and chassis will be discussed together as they are closely dependant on eachother.

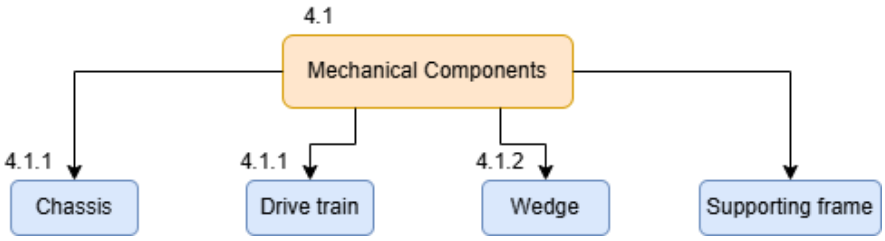


Figure 4.1.1: Mechanical overview of the autonomous sumorobot the numbers in this figure will be updated.

4.1.1 Drive-train and Chassis

This section discusses the two main drive-train concepts that were considered for the autonomous sumorobot. There are various other drive-trains which can be used - tank tracks or caster wheels for example. However, given the size and weight constraints of the Mini Sumo class as well as the chosen strategies of interest, these options were not considered viable.

4.1.1.1 Concept One: Four-Wheel Drive

The four-wheel drive concept uses four independently driven wheels with each wheel controlled by its own motor. Driving all four wheels increases traction, stability and pushing force while distributing the robot's weight more evenly across four contact points. The chassis should keep the motors aligned and position heavier components, such as the battery, low and near the centre to reduce the risk of tipping.

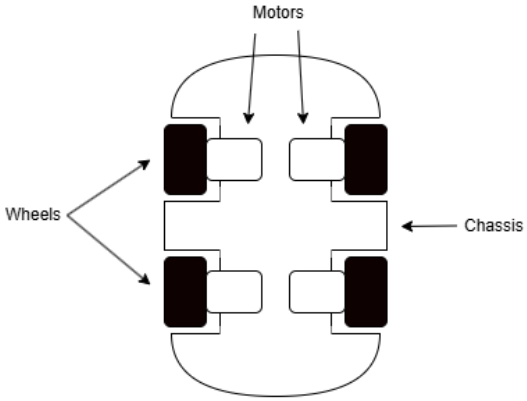


Figure 4.1.2: Four-wheel drive concept.

However, the additional motors increase mass, power consumption, wiring complexity and the space required within the 100 mm × 100 mm size limit. The four-wheel layout also experiences greater turning resistance due to wheel skidding. Overall, this concept prioritises traction, stability, and pushing performance (Arduino Forum, 2013).

4.1.1.2 Concept Two: Differential Drive

Differential drive uses two wheels, each driven by its own motor. This setup is commonly used to ease control of movement and turning as the robot can turn by varying the speed of each wheel. While this design won't reach the same level of traction and pushing force as a four-wheel drive, it is simpler to implement and is commonly used when the goal is to out-maneuver the opponent rather than out-torque them.

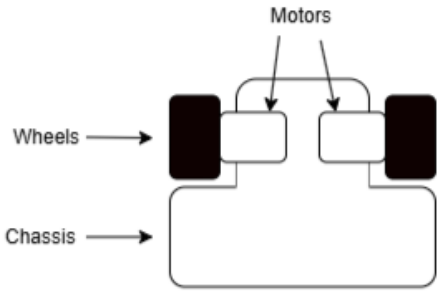


Figure 4.1.3: Differential drive concept.

Differential drive is known for its zero turning radius (Sisko, 2017), which allows the robot to turn in place and quickly change direction. This is particularly useful in a sumo competition where the robot must react quickly to the opponent's movements and navigate within the limited space of the dohyo.

4.1.1.3 Final Concept: Differential Drive

Differential drive was selected as the final concept given the goal to out-maneuver the opponent - especially given limitations faced when trying to find suitable motors. This design allows for either two N20 motors to operate inline with eachother, or, alternatively, two wormgear DC motors can be placed side by side.

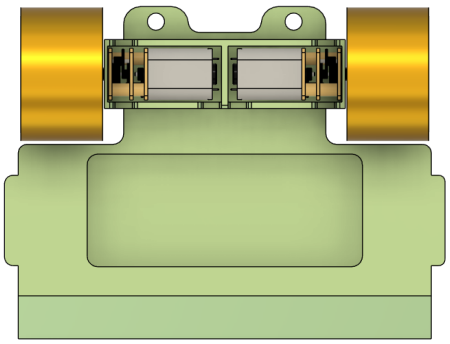


Figure 4.1.4: Final differential drive concept.

The chassis was designed to accommodate the motors, wheel-layout and some

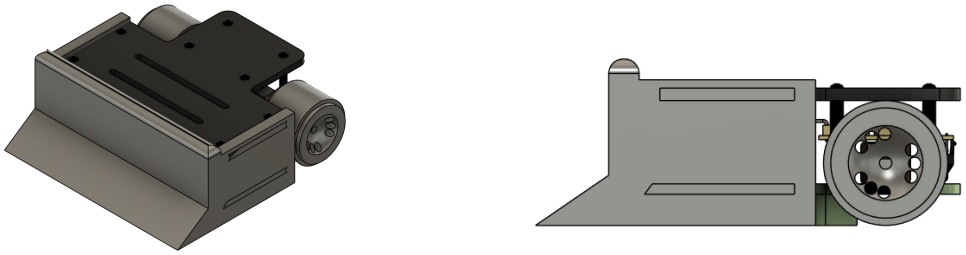
other components such as the battery. These specific accomodations will be discussed in Chapter 5.

4.1.2 Wedge Design

This section discusses the various wedge concepts that were considered for the autonomous sumorobot. There are various other wedge designs that could have been considered - for example, a spring-loaded wedge or a wedge with a lifting mechanism. However, given the size and weight constraints of the Mini Sumo class as well as the chosen strategies of interest, these options were not considered viable and could possibly lead to more areas of weakness rather than strength.

4.1.2.1 Concept One: Flat Wedge

The first wedge concept considered was a flat wedge. This design is simple to manufacture and can be effective at lifting an opponent's wedge if the robot is able to get underneath it. Furthermore, this design allows for more space within the chassis for electrical components as well as room for sensor placement along the flat sides.

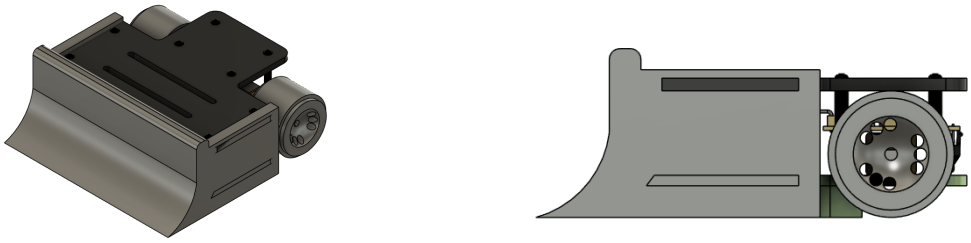


(a) Isometric view (b) Side view

Figure 4.1.5: Conceptual views of the flat-wedge design.

4.1.2.2 Concept Two: Curved Wedge

The curved wedge design is similar to the flat wedge, but with a curved front edge translating into a straight flat surface. This design is intended to help the robot get underneath an opponent's wedge and lift it, while also providing a flat surface for sensor placement.



(a) Isometric view (b) Side view

Figure 4.1.6: Conceptual views of the curved-wedge design.

4.1.2.3 Concept Three: Forked Wedge

The forked wedge design was considered to allow the robot to get underneath an opponent's wedge and lift it, while also providing a flat surface for sensor placement. The forked design allows the robot to straddle an opponent's wedge and lift it from both sides, potentially providing more stability during the lifting process.



(a) Isometric view (b) Front top-down view

Figure 4.1.7: Conceptual views of the forked-wedge design.

4.1.2.4 Final Concept:

DISCLAIMER: for this section, the final wedge design may still include a skirt along the sides - it depends on the sensor placement. The final wedge chosen was a slightly curved wedge with a flat front edge.

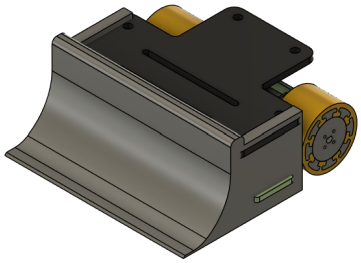


Figure 4.1.8: Final wedge design.

4.2 Electrical Concept Designs

The concepts discussed in this section only focus on a select few electrical components of interest. Components such as the motor drivers and microcontroller will be discussed in the refinement stage once required specifications have been determined. This section mainly discusses the sensors and possible concepts regarding feedback and control systems.

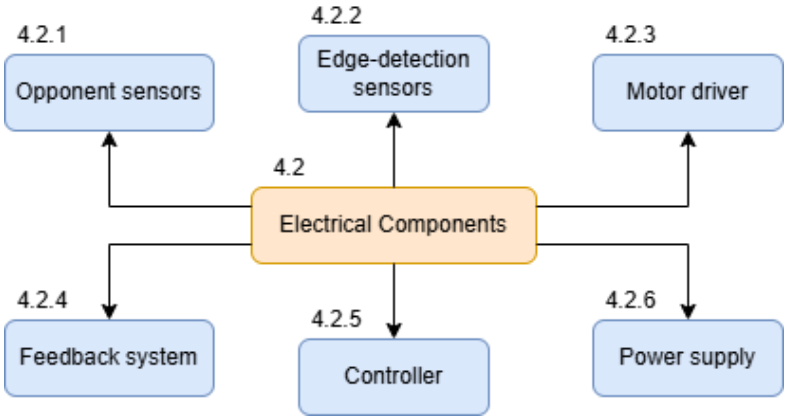


Figure 4.2.1: Electrical overview of the autonomous sumorobot (the numbers in this figure will be updated).

4.2.1 Sensors for Opponent Detection

There are various sensor types that can be used to detect opponents. During the analysis and concept generation stage of which sensors would be the most appli-

cable, it was important to consider uncertainty regarding the opponent's design.

4.2.1.1 Concept One: ToF Ultrasonic Sensors

The first concept considered was ultrasonic sensors - specifically the HC-SR04. These sensors are easy to implement both in hardware and software. Alongside that, they are relatively inexpensive compared to other sensors.

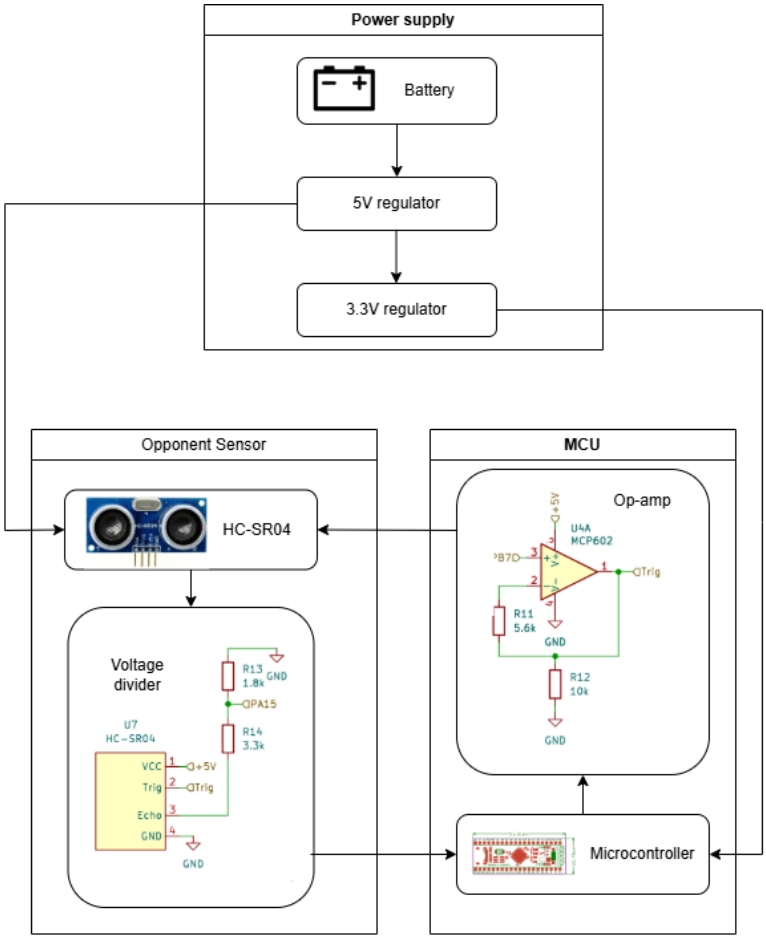


Figure 4.2.2: HC-SR04 ultrasonic sensor implementation.

These sensors work by sending out a pulse of ultrasonic sound and measuring the time it takes for the echo to return. The distance calculation can be implemented into the microcontroller's software through the following equation:

$$\text{Distance} = \frac{\text{Time} \times \text{Speed of Sound}}{2} \tag{4.2.1}$$

The ultrasonic sensor operates at a frequency of 40 kHz which is well below the required sensing latency of 100 ms (TPM_10) (ELECFreaks, n.d.). Alongside this, the sensor has a beam angle of 15–20 degrees and an operational range of 2 cm to 400 cm (ArcBotics, n.d.). Multiple sensors can be used to cover a wider area, allowing the robot to detect opponents from various angles. However, the sensor fails to detect objects that are too close, at an angle or that consists of a soft material. Furthermore, they operate at a voltage of 5 V, which is higher than the operating voltage of most micrcontrollers of interest. To accomodate this, voltage dividers and operational amplifiers would need to be implemented to ensure the sensor can be used properly. Overall, ultrasonic sensors are a good option for detecting opponents, but they have limitations that need to be considered when designing the robot's sensing system.

4.2.1.2 Concept Two: IR Sensors

Another sensor commonly used for object or obstacle detection is the infrared (IR) sensor. These sensors work by emitting an IR light and measuring the amount of light that is reflected back to the sensor. The distance to an object can be calculated based on the intensity of the reflected light.

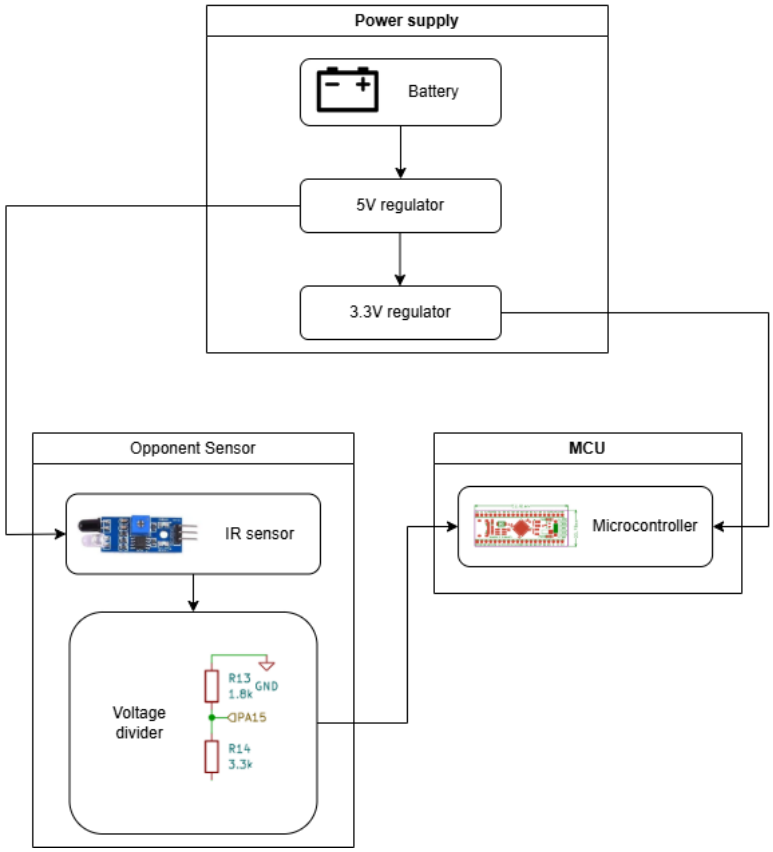


Figure 4.2.3: IR sensor implementation.

IR sensors operate between 3.6 V and 5 V (AM Electronics, n.d.). Similar to the HC-SR04, a voltage divider would need to be implemented to interface the sensor with a microcontroller operating at 3.3 V. The IR sensor has an effective range of 2 cm to 80 cm and has an on-board potentiometer to adjust the sensitivity of the sensor. The sensor has a beam angle of 35 degrees, which is slightly wider than the HC-SR04. However, the IR sensor is more susceptible to interference from ambient light and may not work well in bright environments.

4.2.1.3 Final Concept: ToF Laser Sensors

The final concept that was chosen for opponent detection was the VL53L0X time-of-flight (ToF) laser sensor. This sensor uses a laser to measure the distance to an object by calculating the time it takes for the laser light to reflect back to the sensor.

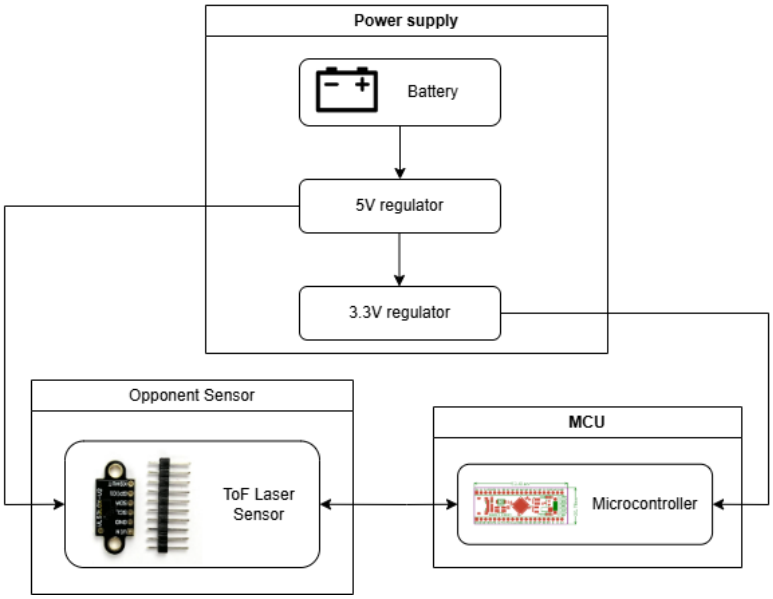


Figure 4.2.4: VL53L0X ToF laser sensor implementation.

These sensors can detect opponents at odd angles and are not as sensitive to ambient light as the IR sensors. The VL53L0X has a range of 30 mm to 2 m and an operating voltage of 2.6 V to 3.5 V (STMicroelectronics, 2024). The sensors operates at a frequency of 50 Hz, which is slower than the other sensors, but still well within the required sensing latency of 100 ms (TPM_10). Given the capability of operating at 3.3 V, the sensor can be directly interfaced with a microcontroller which would simplify the hardware design - especially if multiple sensors are involved. But the simpler hardware design comes at a cost - the VL53L0X is requires I2C communication to interface with the microcontroller. With multiple sensors involved, the registers of each sensor would need to be changed and the I2C bus would need to be managed properly to ensure that the sensors can be read correctly.

Overall, the VL53L0X is a good option for opponent detection and was specifically chosen for detecting opponents with odd geometrical designs as well as its very compact footprint, but it requires careful consideration of the hardware and software design to ensure proper operation.

4.2.2 Dohyo Edge Detection Sensors

During the selection process of these sensors, lane-following sensors were researched. Most sensors that could be used and were available at local suppliers were either too large or far too expensive to be considered for the autonomous sumorobot.

The only viable sensors were already discussed in the previous section, and so the same sensors were considered for dohyo edge detection. Specifically, the infrared (IR) sensors were chosen as they can detect the white edge of the dohyo based on the amount of light reflected compared to the black surface of the dohyo. These sensors have a relatively fast response time which is highly neccessary for the dohyo edge detection task. The only limitation would be the relatively large footprint. However, the solution to this will be discussed during the refinement stage of the design.

4.2.3 Feedback and Control Systems

Most hobby robots do not have any feedback or control systems implemented. This is because they are often designed to perform a specific task in a controlled environment. However, the autonomous sumorobot will be operating in a dynamic environment where the opponent's movements are unpredictable. In the case that an unpredictable attack is made by the opponent, the robot must be able to react quickly and adjust its movement accordingly. This requires a feedback and control system to be implemented.

4.2.3.1 Concept One: Light-interrupter Wheel Encoders

Light-interrupter wheel encoders are a type of feedback system that can be used to measure the speed of the wheels. This is often useful for controlling the speed of the robot and ensuring that it is moving in a straight line. However, light-interrupter wheel encoders require a specific setup to work properly. In this case, they would require a wheel with an extended shaft or a wheel radius large enough to house the encoder.

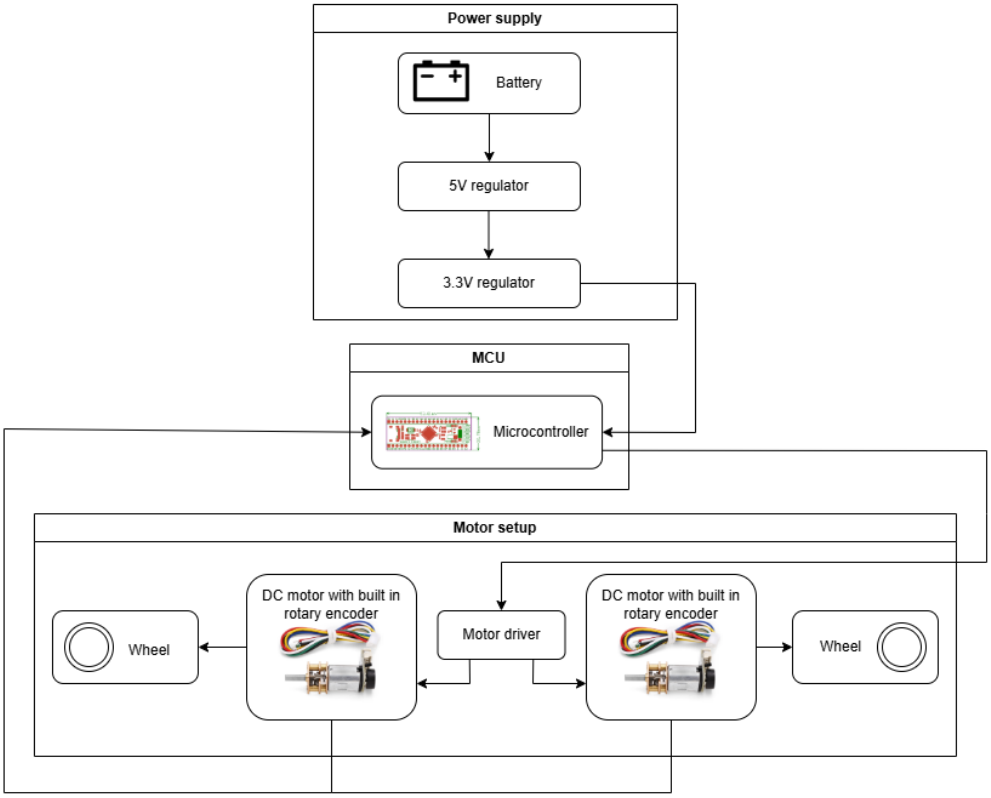


Figure 4.2.5: Light-interrupter wheel encoder implementation.

4.2.3.2 Concept Two: IR-sensor Wheel Encoders

Given the limitations regarding the availability of pre-mounted wheel encoders, the second concept considered was to use IR sensors to detect the rotation of the wheels. This would require a specific setup to work properly, but it would allow for more flexibility in the design of the robot. This concept would require a larger wheel radius to house the IR sensor - increasing speed and decreasing torque.

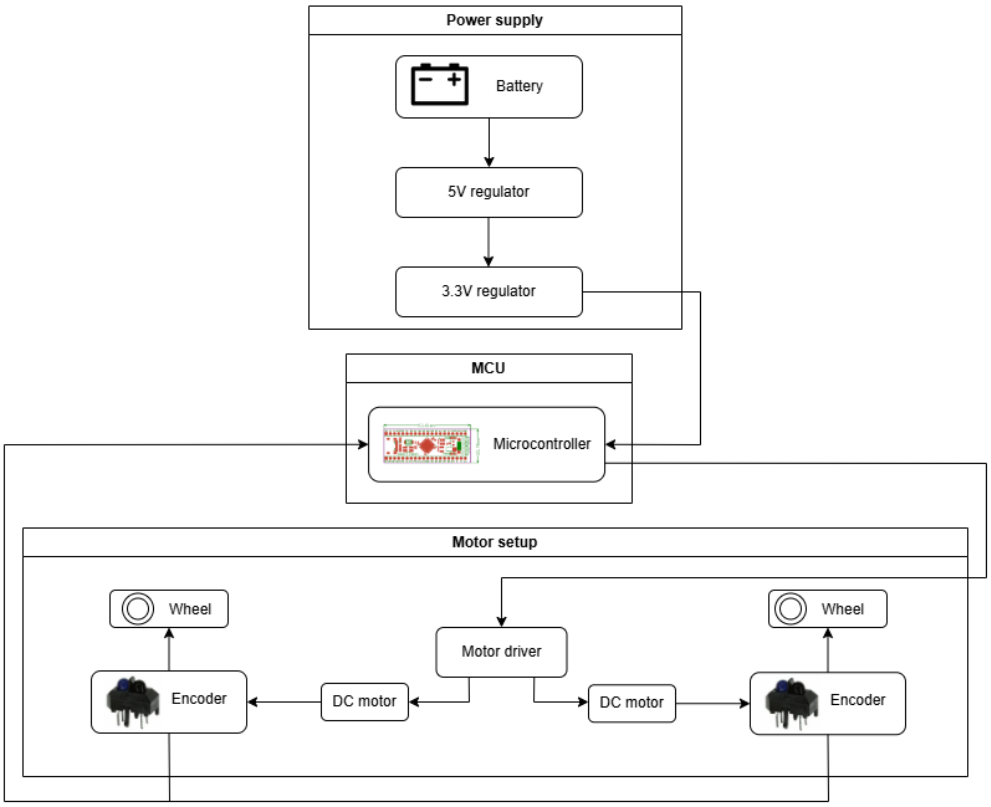


Figure 4.2.6: IR-sensor wheel encoder implementation.

4.2.3.3 Final Concept: MPU-6050 IMU with Potentiometers

In both of the previous concepts, only wheel speed could be measured. However, the final concept considered was to use an MPU-6050 IMU to measure the angular velocity of the robot as a whole. This would allow for more accurate control of the robot's movement and would not require any specific alongside the motors setup to work properly. However, the issue of varying wheel speeds would still need to be addressed. A common solution to this is to use potentiometers to manually tune and adjust the speed of each motor as needed.

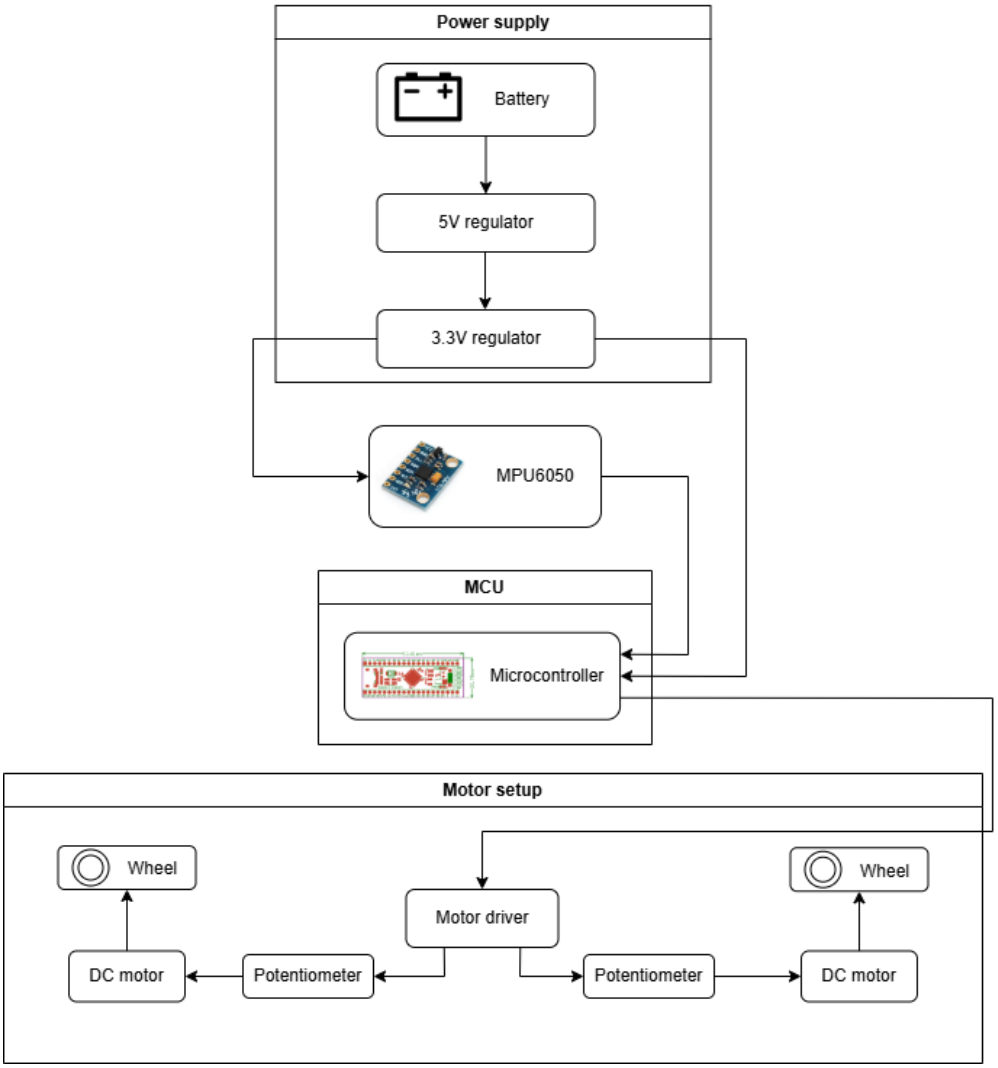


Figure 4.2.7: MPU-6050 IMU implementation.

This approach would then allow the system to meet FR_05 along with the sensors mentioned before.

Chapter 5

Design Refinement and Implementation

5.1 Sensor Placement

Note: The sensor placement will link back to the tests done regarding optimizing the angle/ positioning to cover as much of the arena as possble

5.1.1 Opponent Detection Sensors

5.1.2 Edge Detection Sensors

5.2 Motor Selection

Note: A 300rpm N20 motor will be tested alongside the 500rpm motor in case the 500rpm motor does fail the push and speed tests.

The motor selection process was by far one of the trickiest parts of the design process. The motors had to be small enough to fit within the 10cm x 10cm footprint alongside wheels, a battery and all other components. More importantly, the motors had to be locally available. Most small, high-torque motors are specifically designed for American or European Sumo-bot competitions. Most local alternatives are either too weak or far too big. All options available would make it difficult to find a balance between torque and speed. The most suitable solution was to favour one over the other and adapt the attack strategies accordingly.

A 12V 500rpm N20 metal gear motor was selected as the most suitable option for the robot.



Figure 5.2.1: N20 metal gear motor (Robo Factory, n.d.)

The rated operating speed of 480rpm (Robo Factory, n.d.) would allow the robot to reach a top speed of 0.5 m/s - 0.7 m/s when paired with an outer wheel diameter of 30mm (see Appendix B). This is still fast enough to move from one side of the arena to the other within a second (theoretically meeting TPM_07). It should not be neccessary for the motor to reach this speed during most of the match, as the robot will be moving at a slower speed to avoid overshooting the opponent or the edge of the arena. The trade-off to meet this speed is that the torque of the motor is only 0.8kgfcm per motor. This would result in a rated pushing force of 5N from both motors. However, the stall torque of the motor is 3.2kgfcm per motor (Robo Factory, n.d.). While designing around stall torque is not ideal, these motors do still offer more than enough headroom to push more momentarily.

5.3 Motor Driver Selection

NOTE: I am swapping out this motordriver for a TB67H420FTG Pololu driver. They were originally out of stock. This section will still be kept here for contingengy until I can procure the pololu driver.

The L298 DC Motor Driver was selected as the most suitable option for the robot. It is a dual H-Bridge motor driver that can control the speed and direction of two DC motors independently. The L298 can handle a maximum current of 2A per channel, which is sufficient for the N20 motors used in this design. The L298 also has built-in protection features, such as over-temperature and over-current protection, which helps to prevent damage to the motors and the driver itself (STMicroelectronics, 2023).

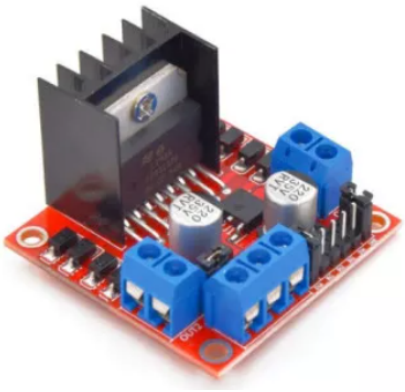


Figure 5.3.1: L298 DC Motor Driver (STMicroelectronics, 2023)

Furthermore, the L298 is widely available and inexpensive compared to most other motor drivers available. However, these drivers are not as efficient as some other options. A maximum voltage drop of roughly 3V can be expected across the driver when the motors are running at full load. Accounting for this, the following realistic torque and speed calculations were made for the motors:

- The maximum voltage that can be supplied to the motors is 8.1V (11.1V - 3V)
- The maximum speed of the motors is 480rpm * (8.1V / 12V) = 324rpm
- The maximum torque of the motors is 0.8kgfcm * (8.1V / 12V) = 0.54kgfcm

This is the absolute maximum voltage drop given a current draw of 2A per motor. In most cases, the current draw would be much less than this (around 300mA per motor).

5.4 Microcontroller Selection

The microcontroller is the central processing unit of the autonomous sumorobot. It is responsible for executing the control algorithms, processing sensor data and controlling the motors. Given its dependance on all other peripherals, the microcontroller could only be chosen at the very end of the design process. During the selection process, there were some other limiting factors as well. The microcontroller had to fit within the 10cm x 10cm footprint of the robot while still having enough pins with the correct functions to control all peripherals of interest. The STM32F411 Blackpill was selected as the most suitable option for the robot. Given its wide variety of pin configurations and compatibility with I2C and SPI, it was

able to meet all the requirements of the robot.

The Blackpill has an ARM Cortex-M4 core running at 100MHz, which provides sufficient processing power for the control algorithms and sensor data processing. It also has 512KB of flash memory and 128KB of SRAM, which is more than enough to store the program code and data. Table 5.4.1 shows the pins needed for each peripheral which aids in the planning of the pin configuration for the microcontroller.

Table 5.4.1: Pin planning for microcontroller

Peripheral	**Pin Type**	**Number of Pins**
VL53L0X Time-of-Flight Sensors	I2C SDA	1
	I2C SCL	1
	GPIO	3–5
IR Edge Detection Sensors	GPIO	3
MPU-6050 IMU	I2C SDA	1
	I2C SCL	1
Motor Drivers	PWM	2
	GPIO	2
Push Button	GPIO	1
DIP Switch	GPIO	4

The Blackpill meets all of these requirements escpecially when taken into account that SDA and SCL lines can be shared between the VL53L0X sensors and the MPU-6050 IMU.

5.5 Power Supply

NOTE: this section might be replaced by a 4S LiPo battery depending on availability

When choosing a power supply, it is important to consider the voltage and current requirements of the entire system. The power supply must be able to provide sufficient voltage and current to all components, including the microcontroller, sensors, motors and any other peripherals.

Lithium polymer (LiPo) batteries are a popular choice for portable electronic devices due to their high energy density and lightweight design. Alongside all of the upsides for using LiPo batteries, it should be noted that they require careful handling and charging to avoid safety hazards. LiPo batteries can be dangerous if punctured, overcharged, or exposed to high temperatures (ChinaHobbyLine, 2024).

With all this in mind, a 3S 11.1V LiPo battery with a capacity of 750mAh and a discharge rate of 45C was selected as the power supply for the system. A datasheet could not be found for this specific component - instead, the specifications provided by the supplier were used to determine the suitability of the battery for the system.

Given the 750mAh and 45C rating of the battery, the maximum current that can be drawn from the battery is 33.75A. Using this and using conservative estimates for all present electrical components, the total current draw of the system was calculated to be 4.4A. This offers a safety margin nearly 8, which is more than sufficient to ensure that the battery can provide enough current to all components without being overdrawn.

5.6 Voltage Regulator and Circuit Protection

Voltage regulators are used to maintain a constant output voltage regardless of changes in input voltage or load conditions. They ensure that the microcontroller and other sensitive components receive a safe and stable voltage supply. The voltage supply was first stepped down to 5V before being further stepped down to 3.3V for the microcontroller and other components. Given the current draw of the system, a LM2596 DC-DC buck converter was selected as the voltage regulator for the system. The LM2596 is a popular choice for voltage regulation due to its high efficiency, low dropout voltage, and wide input voltage range. The LM2596 can handle a maximum input voltage of 18V and can provide a regulated output voltage of 1-17V with an adjustable potentiometer. The LM2596 can provide a maximum output current of 2A (Micro Robotics, 2026), which is sufficient for the current requirements of the sensors and other peripherals (see Appendix B).



Figure 5.6.1: LM2596 DC-DC Buck Converter (Micro Robotics, 2026)

While the voltage regulator is the biggest source of safety for the system regarding voltage fluctations, current protection is also important. A 200 - 300 mA fuse will be installed in series with the power supply to the microcontroller to protect all important components from overcurrent situations. Ideally, the microcontroller should not pull more than 10 - 20 mA while its GPIO pins are rated for 8 mA (STMicroelectronics, 2025).

5.7 Wheel Selection

Most off-the-shelf wheels are made from rubber or plastic or they do not fit within the geometrical constraints of the robot. Rubber or plastic wheels are not ideal given their low coefficient of friction which would result in poor traction and wheel slippage. Molding wheels from a high-friction material such as polyurethane is common practice amongst competitive robotics. Coupled with a larger width, this would provide the robot with a larger contact patch and far better traction than any off the shelf wheels. To achieve this, a wheel mold was designed as shown in Figure 5.7.1. This design was 3D printed alongside the mold cover and wheel hubs.

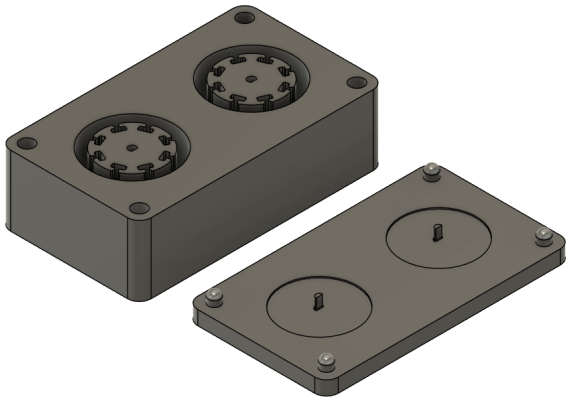


Figure 5.7.1: Wheel Mold

Shore 30 A Vytaflex was selected specifically for its hardness rating. However, this material fuses easily with PLA. While it is ideal for the polyurethane to fuse to the wheel hub, a release agent paired with a draft angle of 5°was used to ensure that the polyurethane would not fuse to the mold cavity. Initially, there were some concerns about keeping the hub in place or preventing it from floating. The density of PLA is 1.25g/cm^3 while the density of Vytaflex is 1.15g/cm^3. Since the density of PLA is higher than that of Vytaflex, the hub would sink into the mold cavity and floating should theoretically not be an issue. However, to ensure that the hub would stay completely in place, a cover was designed with a small extrusion to slot within the center of the hub.

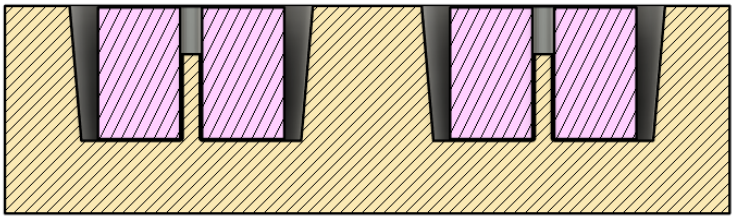


Figure 5.7.2: Wheel Mold Draft Angle

5.8 Frame, Chassis and Wedge Implementation

5.8.1 Frame and Wedge

The frame and wedge are interlinked. The final version will be discussed here once it accommodates the correct sensor placement.

The use of PETG and sheetmetal will be used to account for FR_13 and FR_11 (still to be discussed in more detail).

5.8.2 Chassis

The chassis currently houses the motors and battery. The final chassis will include housing/brackets for the IR distance sensors. This section will also discuss how the chassis interfaces with the frame and why PLA was used insted of PETG/TPU.

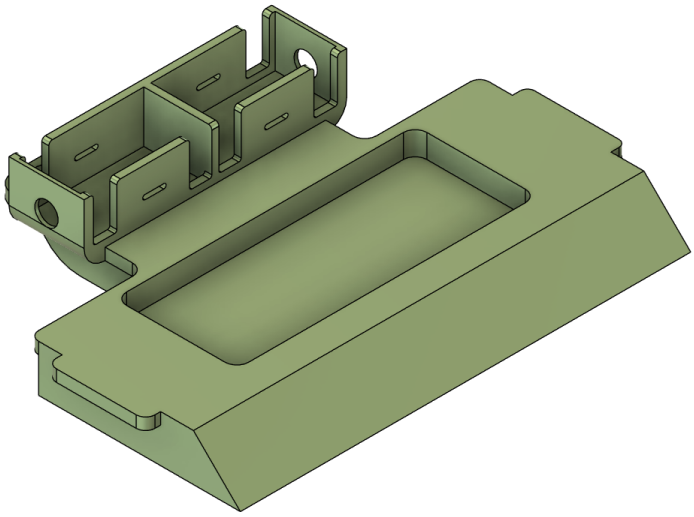


Figure 5.8.1: Chassis

5.9 Summary of Design Decisions

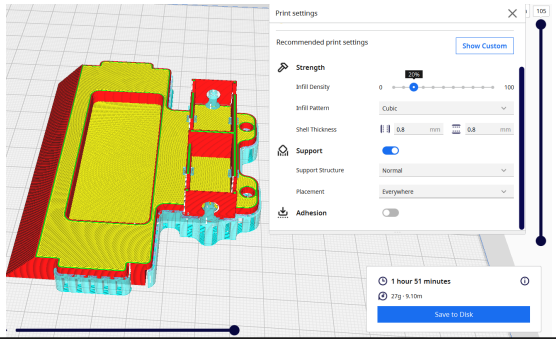
This sub-section has been added to document whether all of the functional requirements have been met. The table below will be merged into discussion for the according sections of the final report.

Table 5.9.1: Functional requirement verification

FR ID	**Verification / How the requirement was met**
FR_01	VL53L0X time-of-flight sensors will be used to detect opponents accurately within the range of the dohyo.
FR_02	IR sensors will be used for edge detection as they can distinguish between different surface colours. IR sensors accurately distinguish between white and black surfaces based on the amount of reflected infrared light.
FR_03	Two N20 motors coupled with an 11.1 V GooSky battery will drive the robot according to the finite state machine implemented within the control software.
FR_04	The selected N20 motors provide sufficient torque, making wheel traction the limiting factor (see Appendix B). Final verification will be completed during physical testing.
FR_05	Combining the MPU-6050 with the selected sensors enables the robot to respond to its opponent and environment in real time.
FR_06	The selected sensors together with the control strategy implemented on the STM32F411 Black Pill allow the robot to operate autonomously.

Continued on next page

Table 5.9.1 continued from previous page

FR ID	Verification / How the requirement was met
FR_07	During modelling, the 100 mm × 100 mm footprint was applied as a strict design constraint. Furthermore, the mass of every component, including the 3D-printed parts, was tracked throughout the design process. 
FR_08	All components are self-contained and require no external power source or communication during operation. Although the GooSky battery is charged externally, no external inputs are required once the battery is installed.
FR_09	The robot not only detects the arena edge but prioritises avoiding it over all other actions. This behaviour is implemented through the finite state machine using high-priority interrupts on the STM32F411 Black Pill.
FR_10	A push-button is incorporated alongside the main power switch. Pressing the button sets a flag, after which the finite state machine begins execution following the mandatory start delay.
FR_11	The robot incorporates no weaponry or external substances that could damage the opponent or increase traction. A master power (kill) switch further improves operational safety.
FR_12	Many preferred components were only available internationally. Consequently, locally available alternatives and practical workarounds were considered throughout the design process.

Continued on next page

Table 5.9.1 continued from previous page

FR ID	**Verification / How the requirement was met**
FR_13	The combination of PETG, PLA, TPU and sheet metal is expected to provide sufficient structural strength to withstand the impacts and pushing forces experienced during Mini Sumo matches.

Chapter 6

Software Design and implementation

Flow-diagrams and code implementation will be discussed here.

6.1 Startup Sequence

6.2 Search Algorithm

6.3 Attack Algorithm

6.4 Back-off Algorithm

6.5 Feedback and Control Systems

Chapter 7

Testing and Refinement

Currently IR sensors have been tested and tuned

Motor driver has been tested and a simple matlab program has been written to help predict tragectory based on different PWM values. This will be used to help tune the wheel speeds/ controller. Sensor input will be added to the existing simulation lateron.

ToF sensors have been tested, data still needs to be mapped in matlab to visualize the scope covered.

Blackpill has been tested with some simple code to test the PWM and GPIO pins/ register allocation for the ToF sensors.

Chapter 8

Conclusions and Future Work

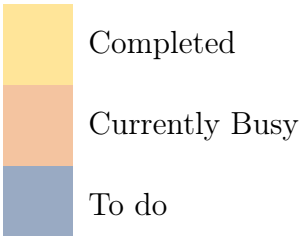
Appendix A

Planning

A.1 Gannt Chart: Progress to Date

Table A.1.1: Project schedule

Task	**Feb 2026**	**Mar 2026**	**Apr 2026**	**May 2026**	**Jun 2026**	**Jul 2026**	**Aug 2026**	**Sep 2026**	**Oct 2026**	**Nov 2026**
Review Literature										
Identify Design Approaches										
Compile Design Requirements										
Investigate Design Concepts										
Select Preferred Design										
Refine the Selected Design										
Simulate Components of Interest										
Finalize Design for Prototyping										
Manufacture Prototype										
Test Prototype										
Improve Prototype and Validate Performance										
Finalize Report										
Presentation										



*Note, the literature review will be updated and expanded in parallel with the design and testing phase

A.2 Budget and Procurement

Table A.2.1: Bill of Materials Procurement Status

Item	Qty	Cost	Total	Found	Email	Proc.	Rec.
Sensors							
ToF	5	R73.91	R369.55	✓	✓	✓	✓
IR	1	R36.00	R36.00	✓	✓	✓	✓
MPU-6050	1	R10.00	R10.00	✓	✓	✓	✓
Shipping + VAT		R208.32	R208.32				
		Sub-total	**R623.87**				
Microcontroller							
STM32F411 Black Pill	1	R158.00	R158.00	✓	✓	✓	✓
Shipping		–	–				
		Sub-total	**R158.00**				
Power System							
N20 DC Motor	2	R35.00	R70.00	✓	✓	✓	✓
Motor Driver	1	R206.00	R206.00	✓	✓	✓	✓
11.1–16 V LiPo Battery	1	R619.00	R619.00	✓	✓	✓	✓
Shipping		R0.00	R0.00				
		Sub-total	**R895.00**				
Miscellaneous							
Prototyping Board	1	R200	R200	✓	✓	✓	✓
DIP Switch	1	R10	R10	✓	✓	✓	✓
3D Printing Frame Parts	1	R1000	R1000	✓	✓		
Power Switch	1	R10	R10	✓	✓	✓	✓
Wheel Material (Vytaflex)	1	R350	R350	✓	✓	✓	✓
Passive Components	1	R400	R400	✓	✓	✓	✓
		Sub-total	**R1970.00**				
		Grand Total	**R3646.87**				

*Man hours will be updated for a separate BOM. The current list shows the progress of procurement/ budget use.

Appendix B

Calculations

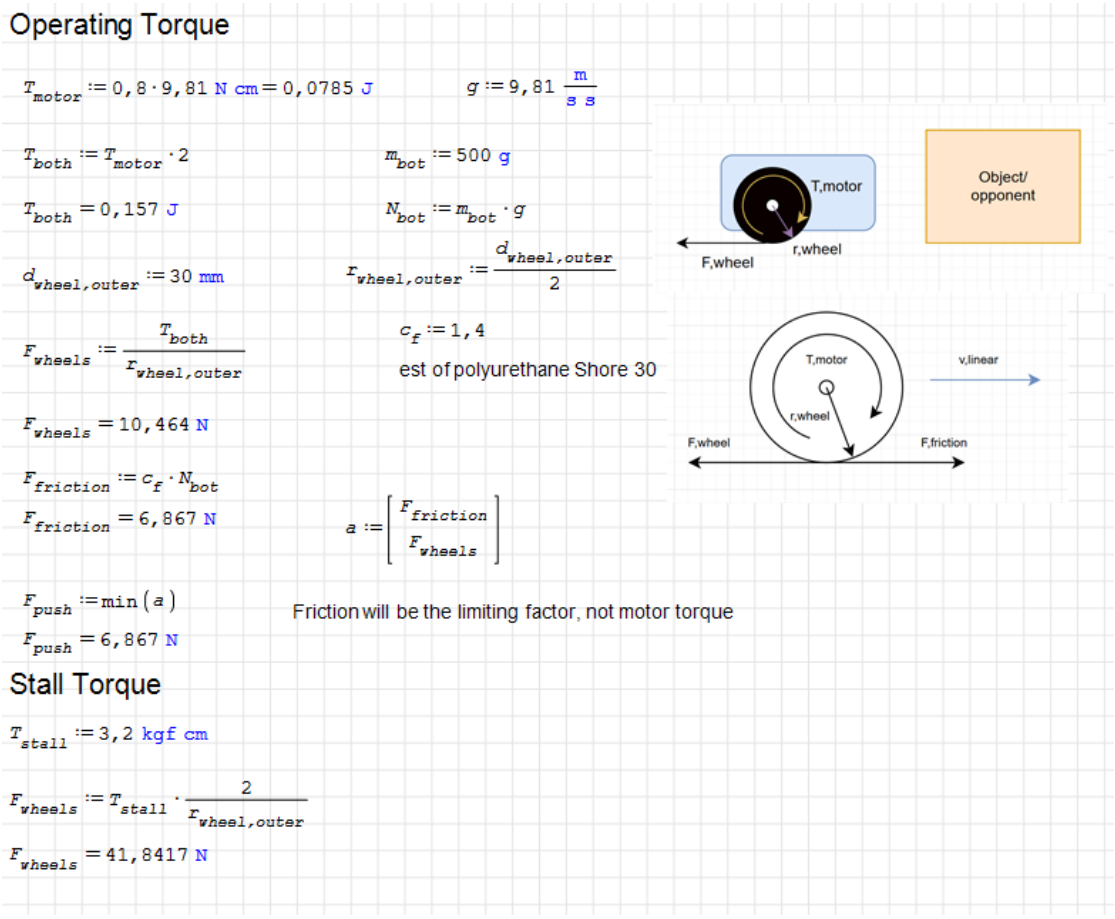


Figure B.0.1: Motor Torque and Wheel Material.

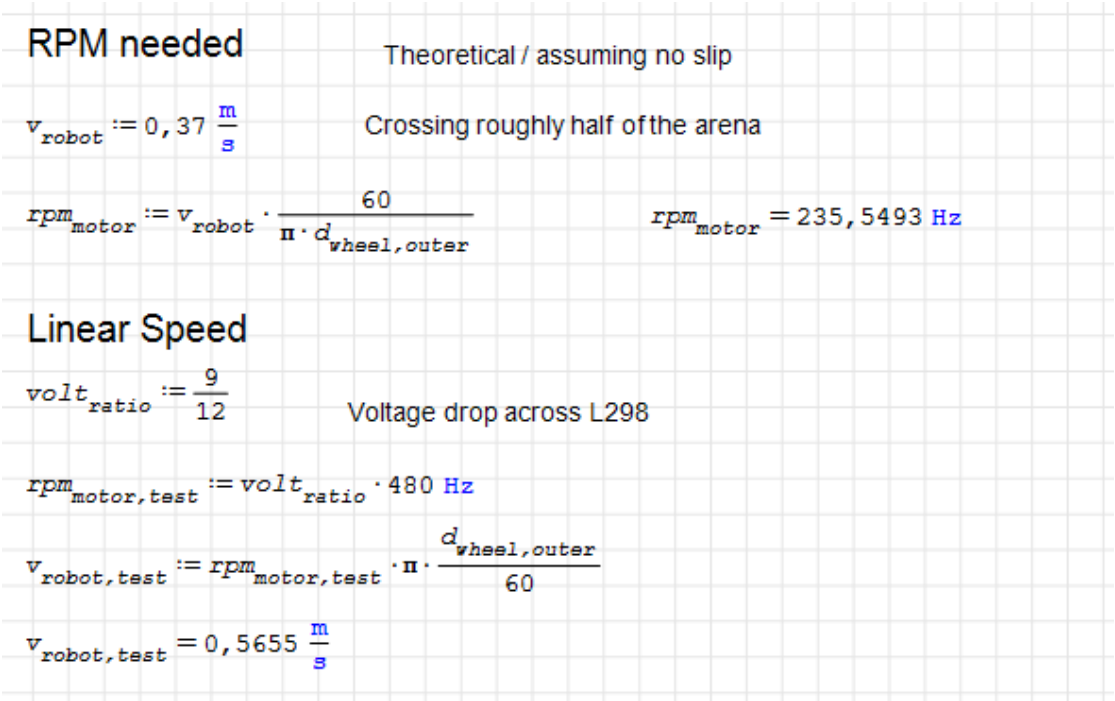


Figure B.0.2: Motor Selection Calculations.

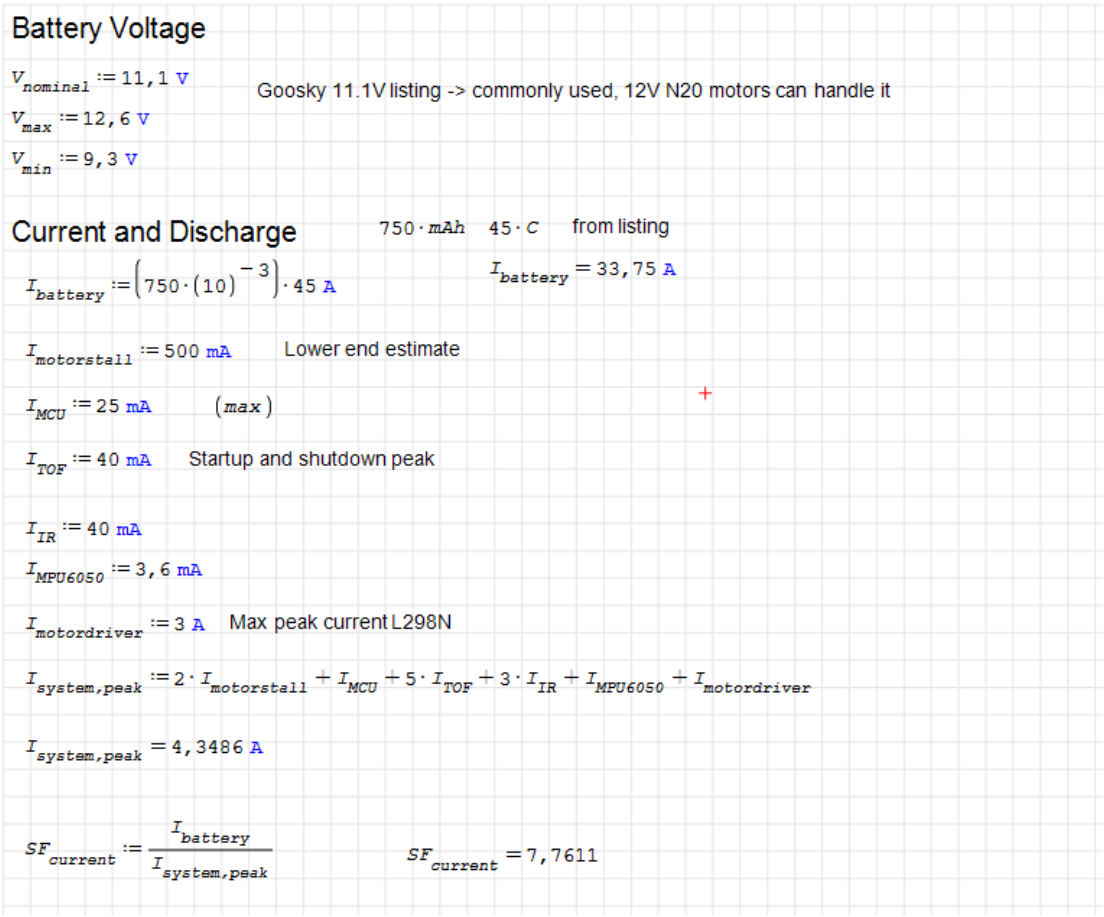


Figure B.0.3: Battery Calculations.

Appendix C

Schematic

C.1 KiCad Schematic

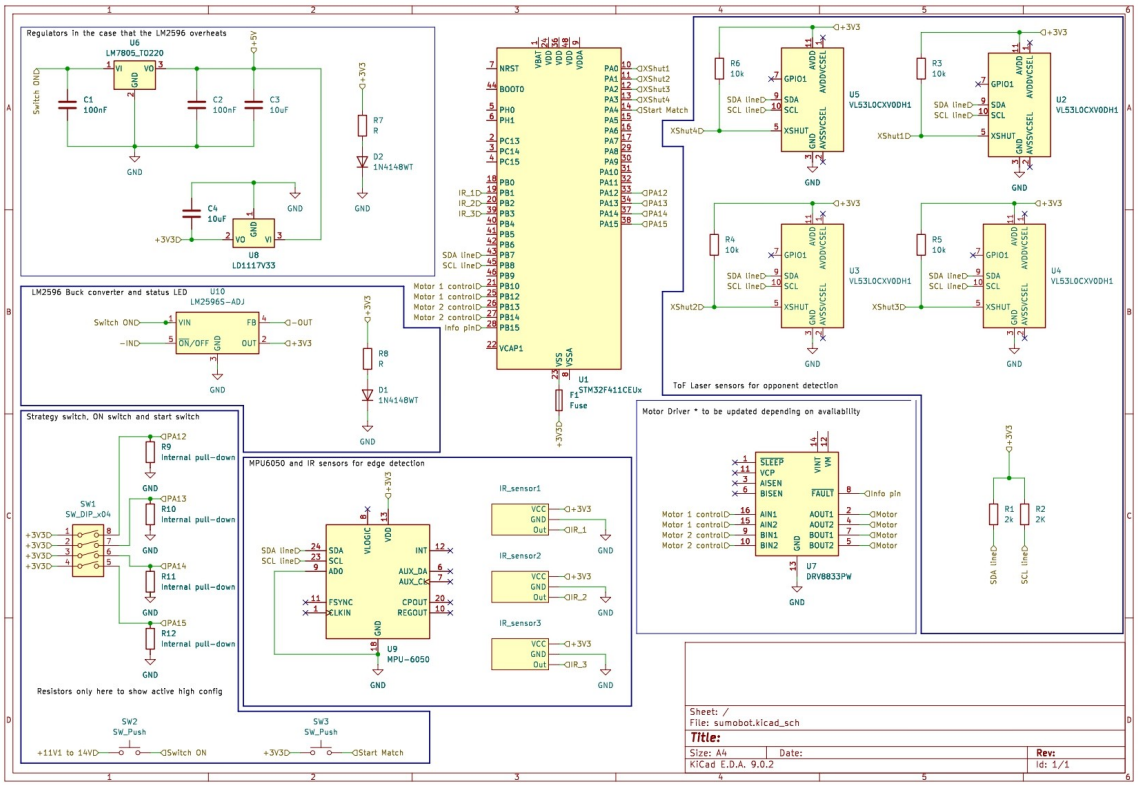


Figure C.1.1: KiCad schematic of the Mini Sumo robot.

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